

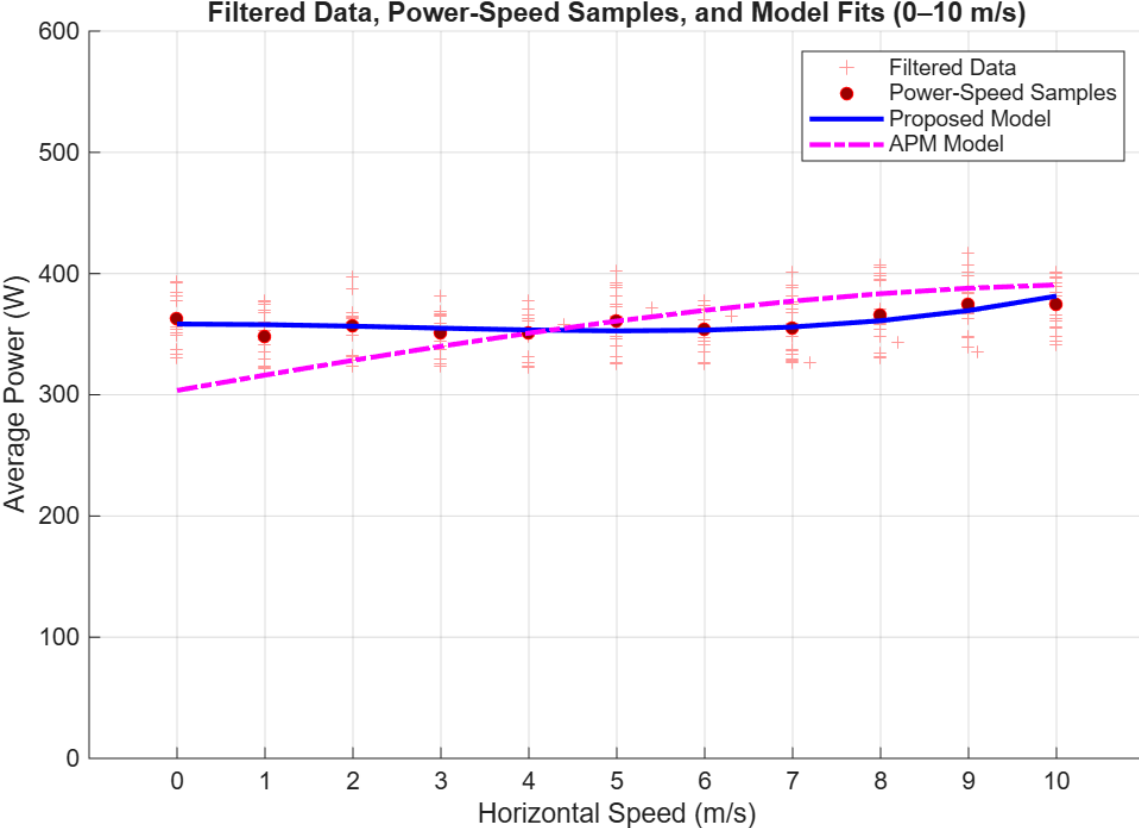
Modeling power consumption of multirotor UAV Thesis progress

Results obtained in case of Horizontal flight

- The paper has **taken a predesigned power model for a single rotor**.
- Then, they **summed the contribution of all rotors** to obtain the total UAV power.
- A key assumption was that the **weight is equally distributed** across rotors, i.e. W/n .
- The thrust of the i -th rotor is denoted as T_i , with total thrust $T = \sum_i T_i$.
- Taken the data In the **horizontal flight case**, data was collected from **0–10 m/s** at an altitude of **20 m**.
- In the **vertical ascent case**, data was taken from **0–5 m/s** at **150 m altitude**, with height constraints ensuring the UAV did not exceed 150 m.
- In the **vertical descent case**, data was recorded from **0–3.5 m/s**, also under the same **150 m altitude constraint**.

TABLE I
List of Main Notations

Notation	Physical meaning	Simulation value
δ	Profile drag coefficient	0.011
ρ	Air density in kg/m^3	1.168
s	Rotor solidity	0.045
A	Rotor disc area in m^2	0.214
Ω	Blade angular velocity in radians/second	-
R	Rotor radius in meter (m)	0.26
k	Incremental correction factor to induced power	0.11
W	UAV weight in Newton	20
$\tilde{\kappa}$	Thrust-to-weight ratio, $\tilde{\kappa} \triangleq \frac{T}{W}$	1
v_0	Mean rotor induced velocity in hover	6.325
$S_{FP\parallel}$	Fuselage equivalent flat plate area in horizontal status in m^2	0.009
$S_{FP\perp}$	Fuselage equivalent flat plate area in vertical status in m^2	0.377
C_T	Thrust coefficient	0.001195

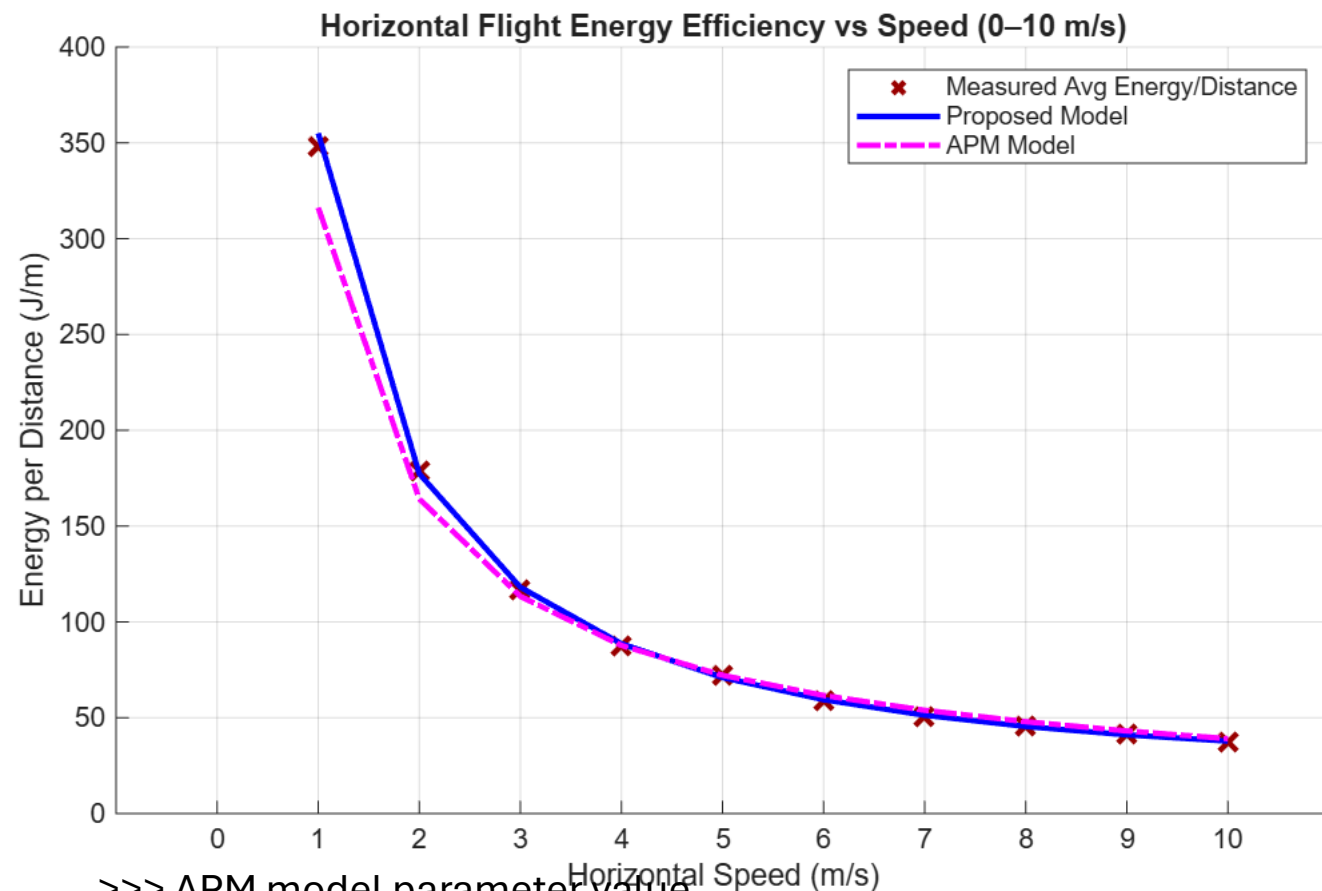


>> proposed model parameter value

Parameter	FittedValue
{'δ'}	0.018
{'s'}	0.047
{'C_T'}	0.0012841
{'k'}	0.11
{'S_fp_parallel'}	0.022797
{'v0'}	6

Average Relative Difference (Proposed Model): 1.24 %

Hover Relative Difference (Proposed Model): 1.26 %

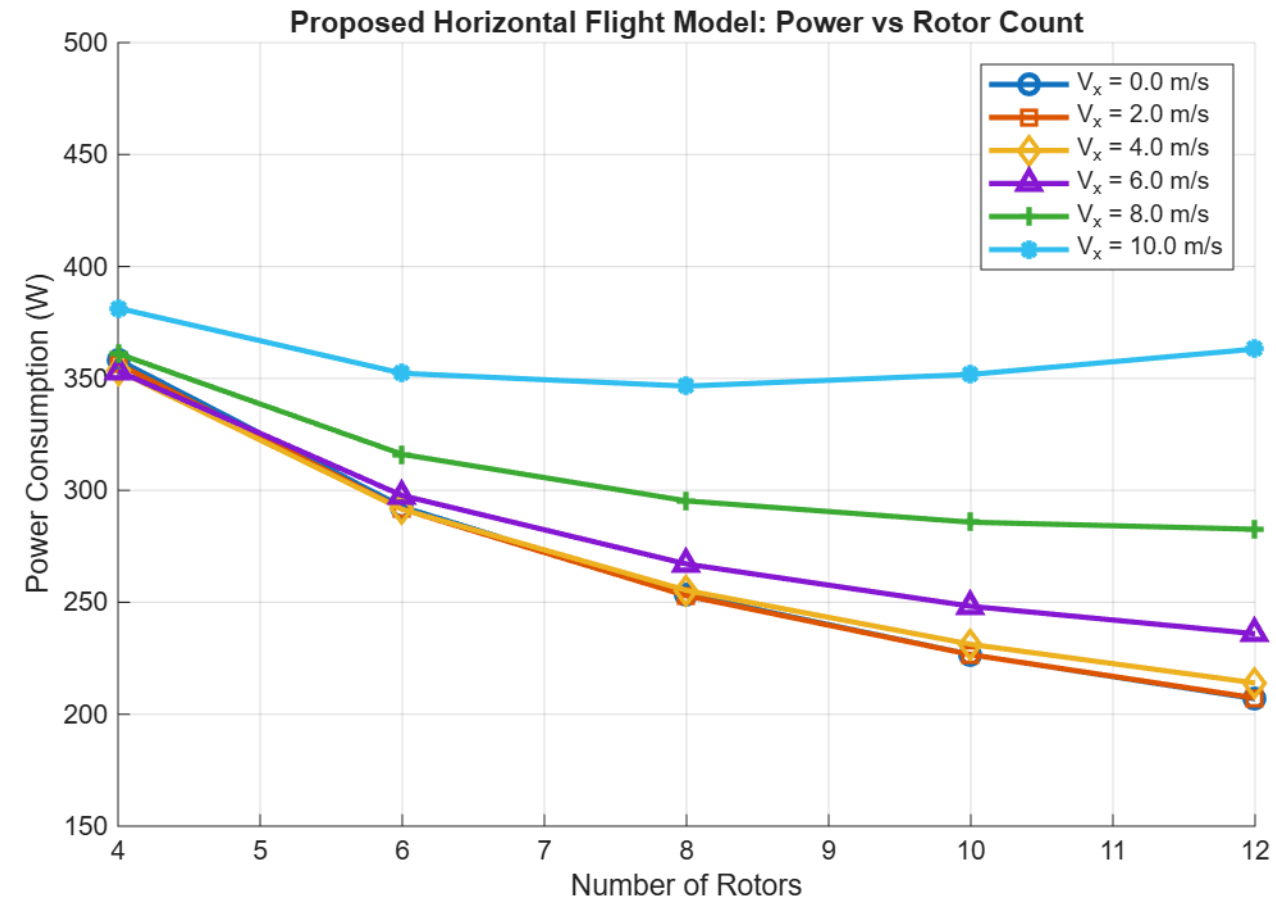
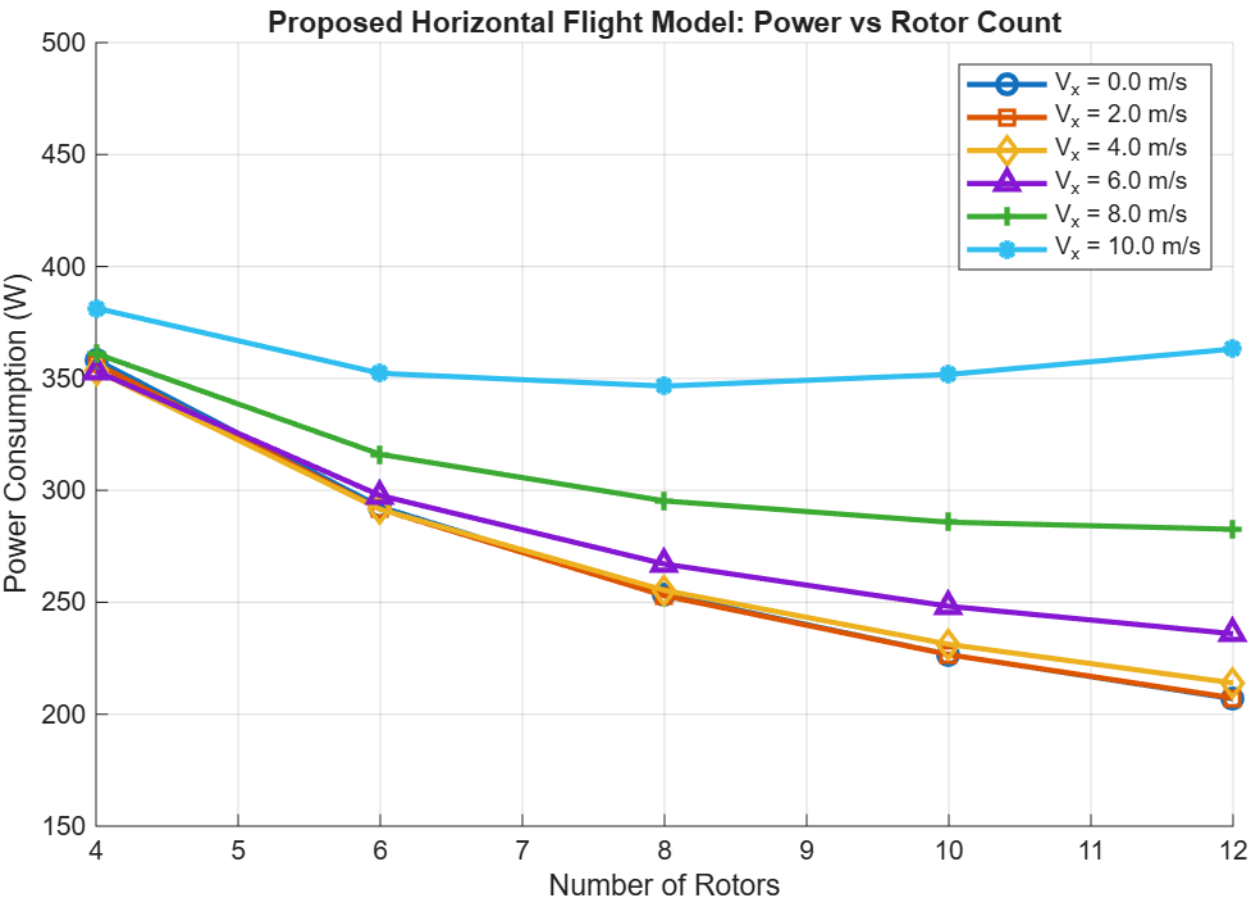


>>> APM model parameter value

Parameter	FittedValue
{'k1'}	0.87537
{'c2'}	1
{'c4'}	0.01
{'c5'}	0.05

Average Relative Difference (APM Model): 5.46 %

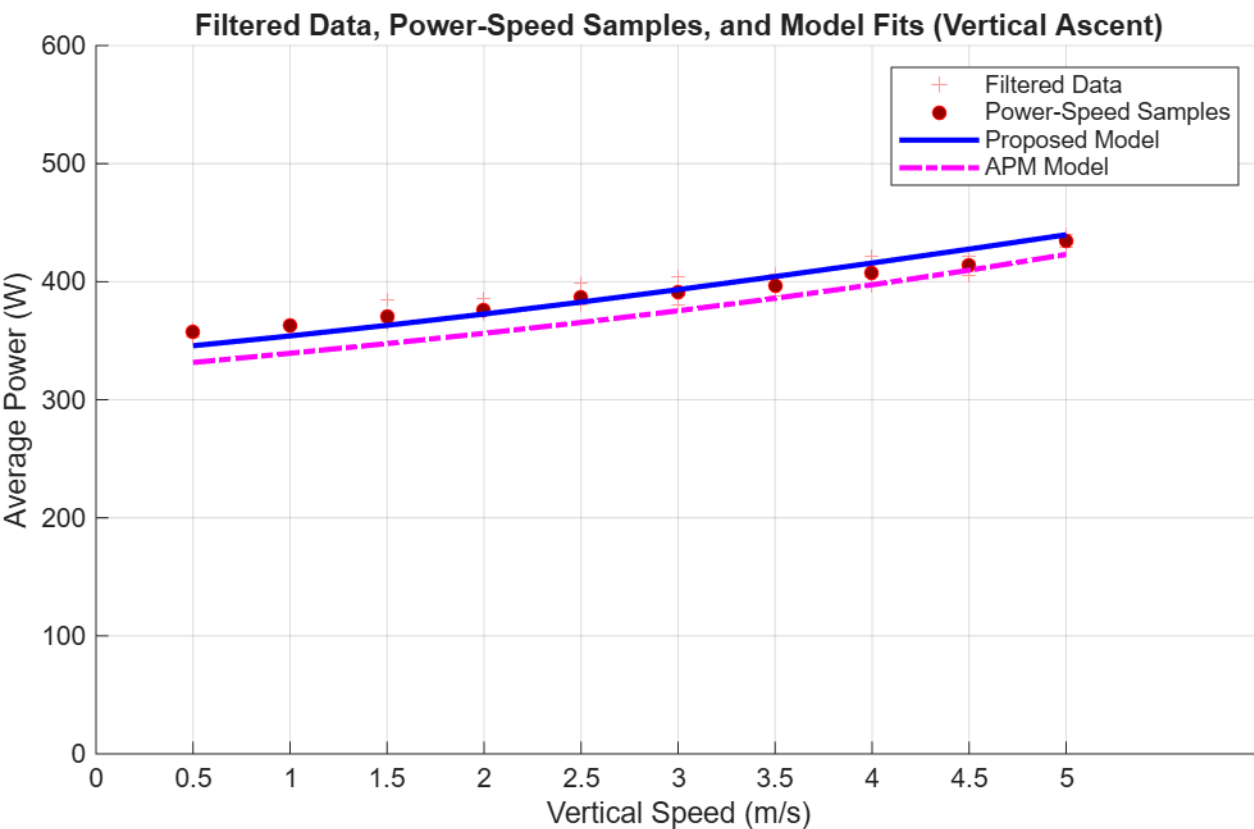
Hover Relative Difference (APM Model): 16.36 %



$V_x = 0.0$ m/s: Power change from $n=4$ to $n=12$ = -42.26 % (decrease)

$V_x = 0.0$ m/s: Power change from 0.1kg to 1.0kg payload = 46.57 % (increase)

Results in case of vertical ascent

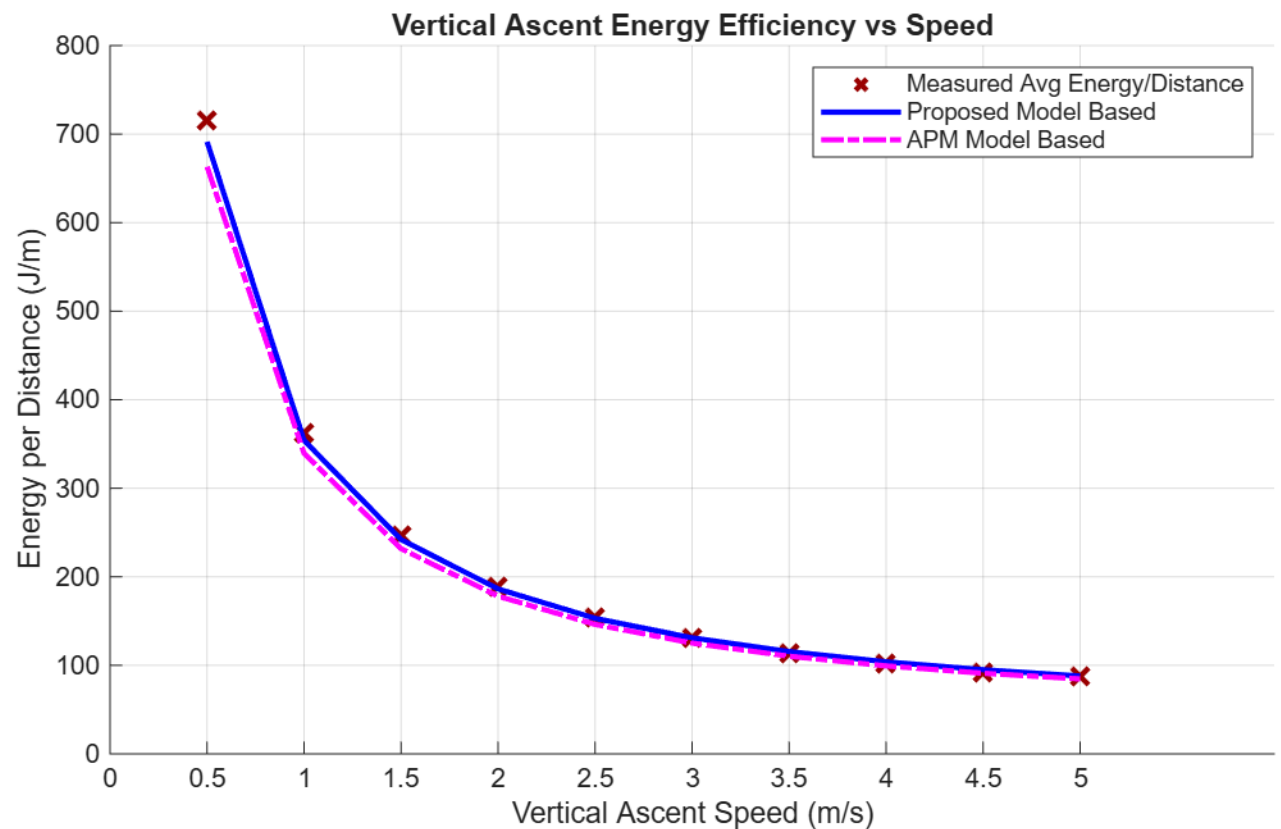


>> proposed model parameter value

Parameter	FittedValue

{'δ' } 0.010827
{ 's' } 0.044281
{ 'C_T' } 0.001223
{ 'k' } 0.11416
{ 'S_fp_perp' } 0.009

Average Relative Difference (Proposed Model): 2.14 %

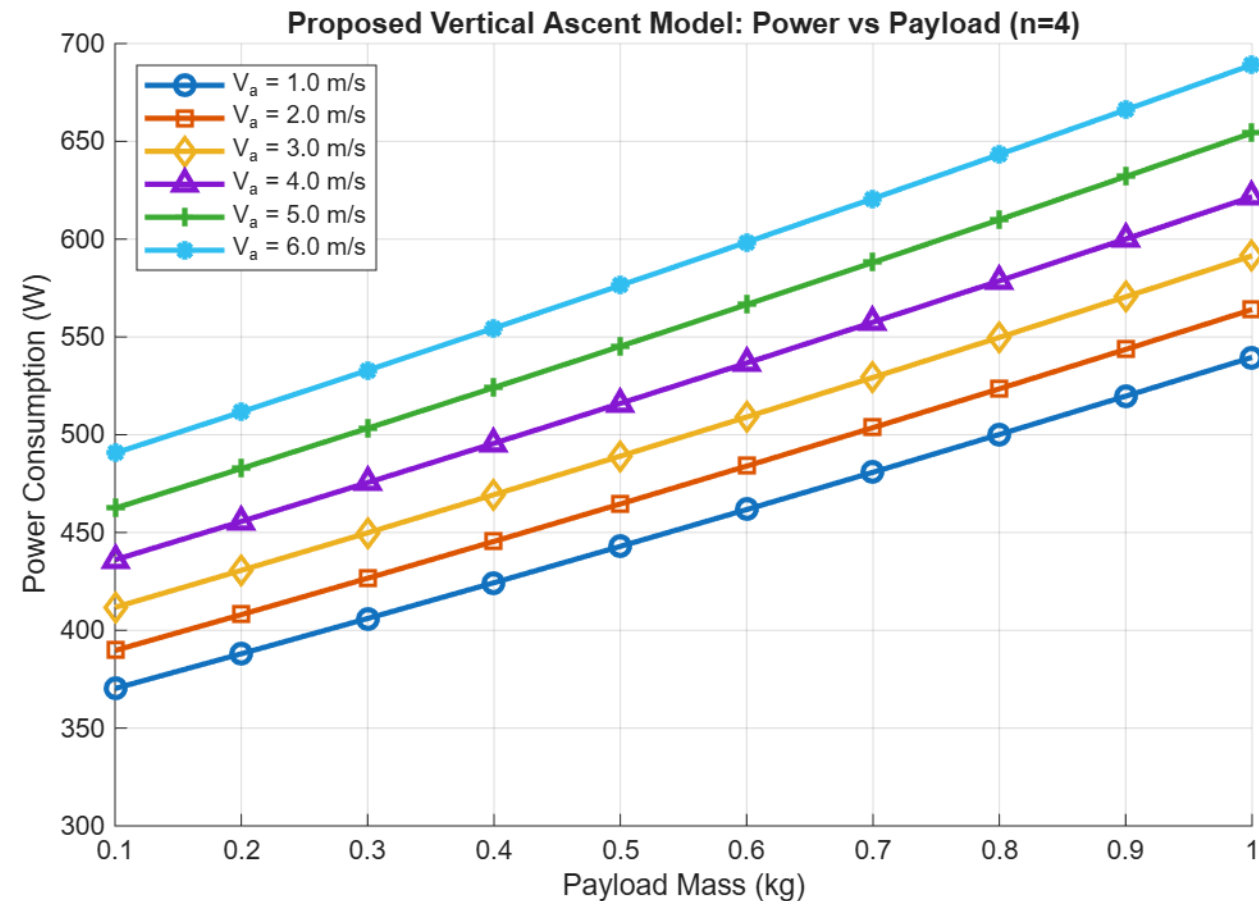
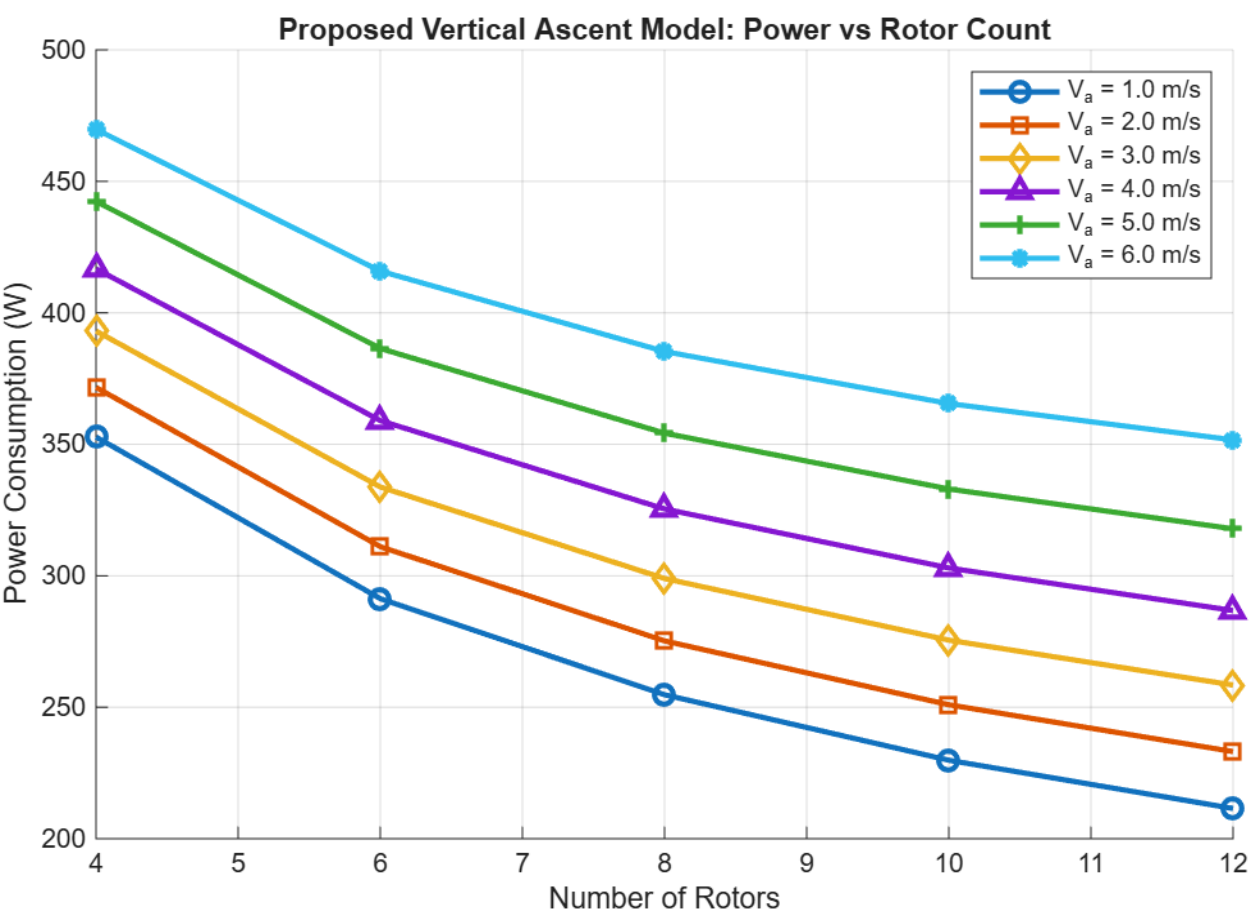


>>> APM model parameter value

Parameter	FittedValue

{ 'k1' } 1
{ 'c2' } 1
{ 'c4' } 0.1
{ 'c5' } 0.01

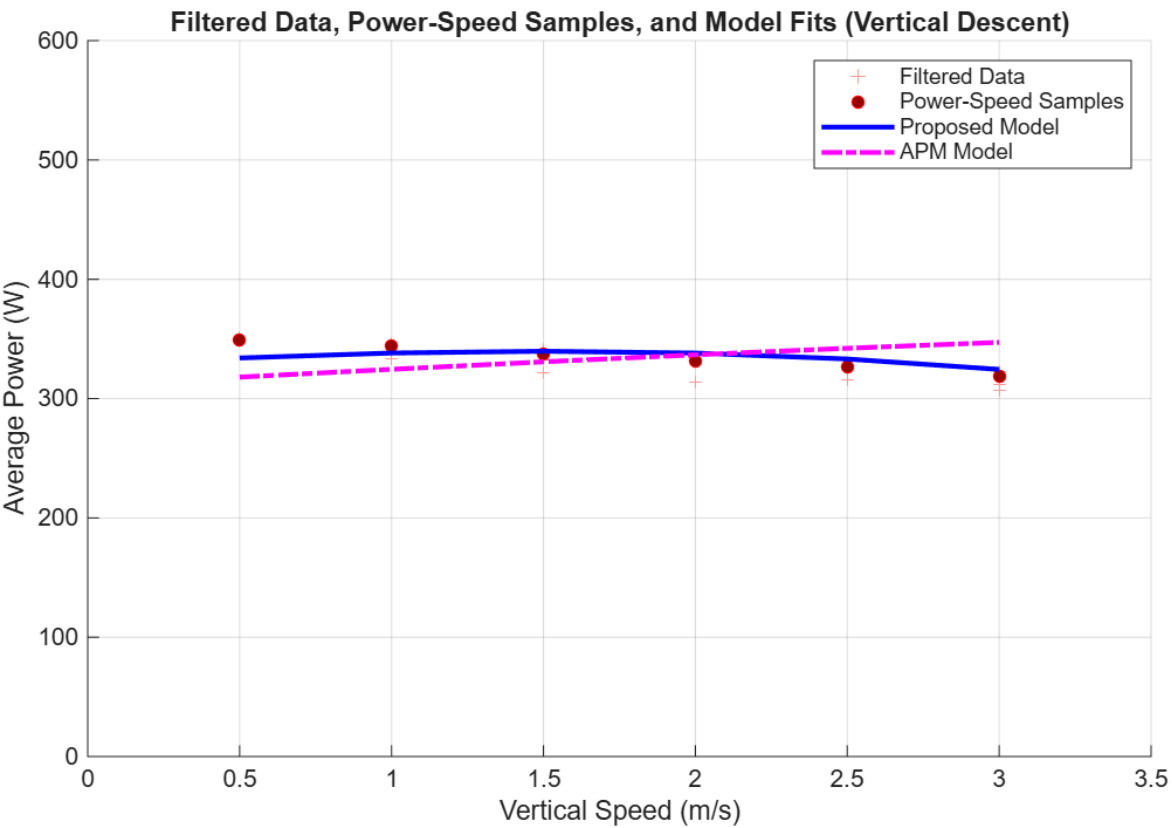
Average Relative Difference (APM Model, mg & k2 fixed): 4.39 %



$V_a = 1.0$ m/s: Power change from $n=4$ to $n=12$ = -40.05 % (decrease)

$V_a = 1.0$ m/s: Power change from 0.1kg to 1.0kg payload = 45.72 % (increase)

Results in case of vertical descent

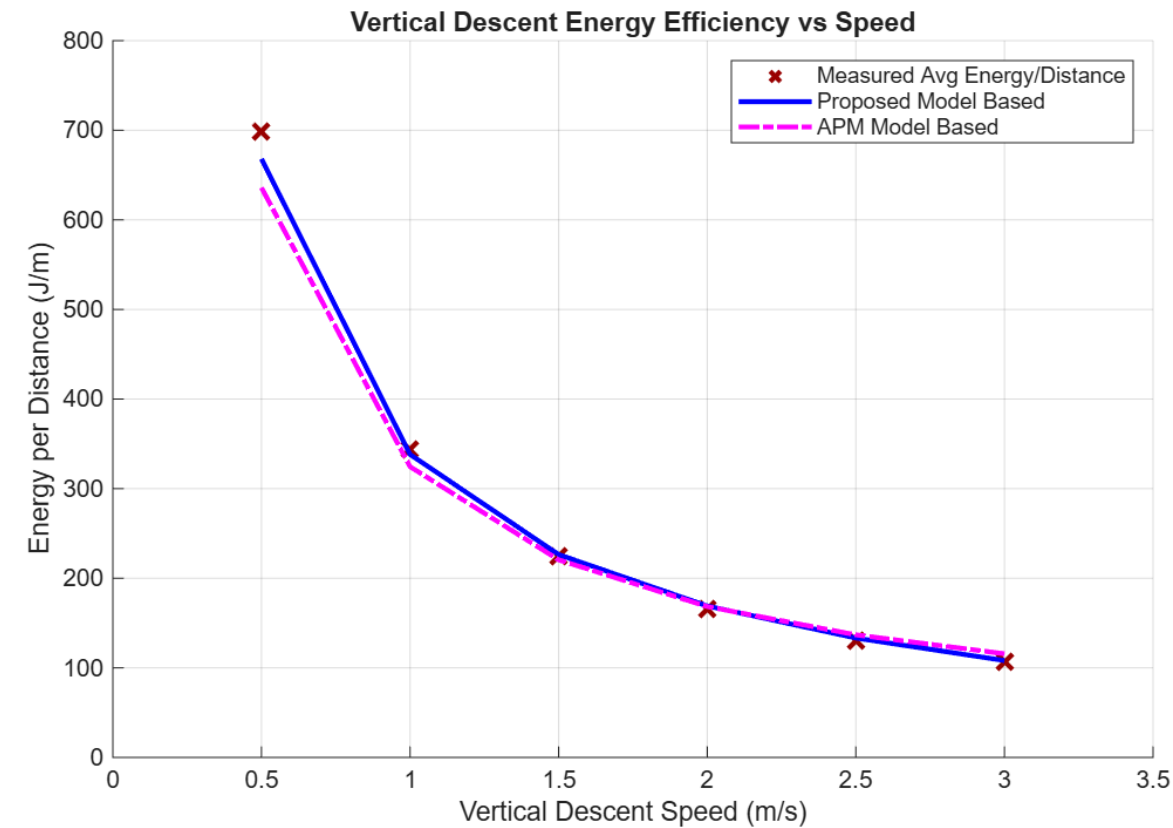


>> proposed model parameter value

Parameter	FittedValue

```
{'δ' } 0.010674
{'s' } 0.043648
{'C_T' } 0.0012435
{'k' } 0.10813
{'S_fp_perp'} 0.48
```

Average Relative Difference (Proposed Descent Model): 2.03 %

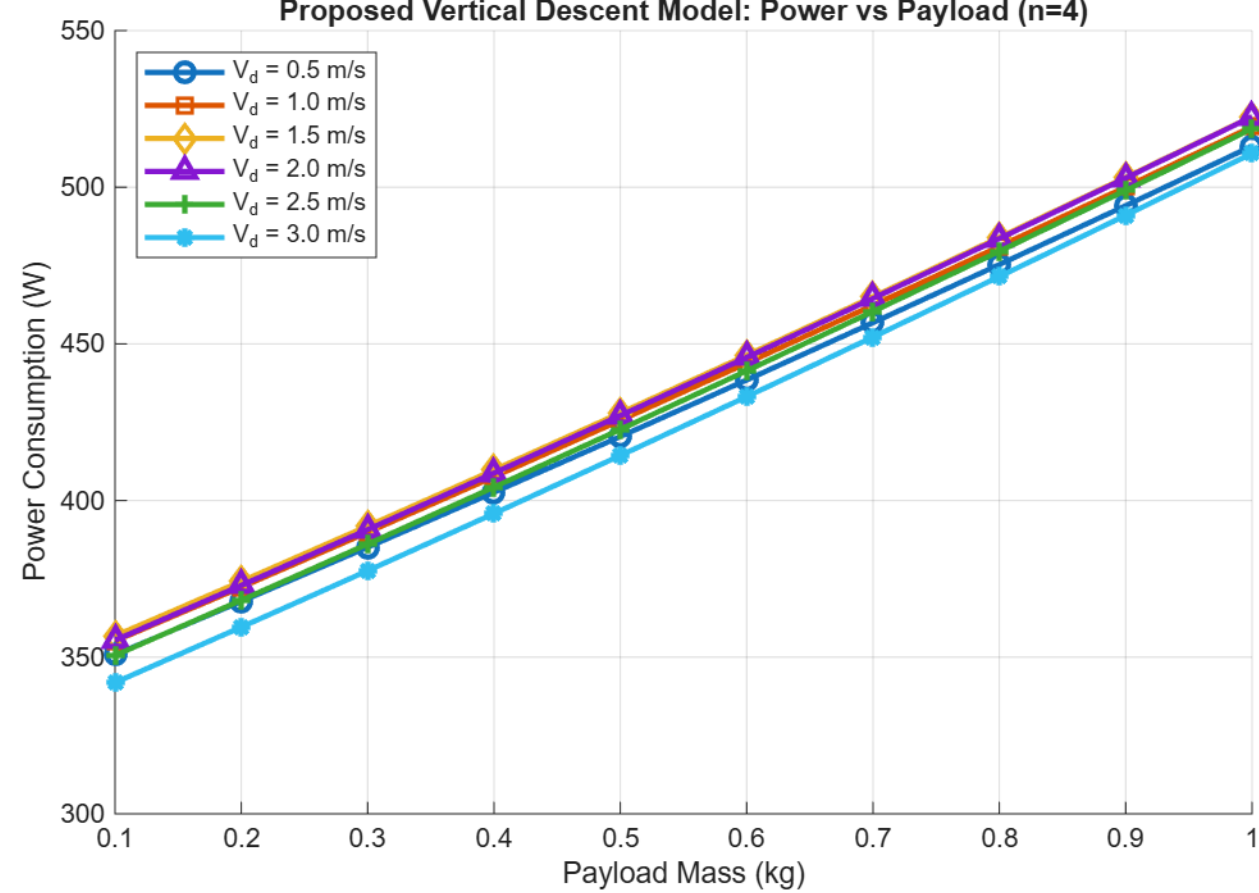
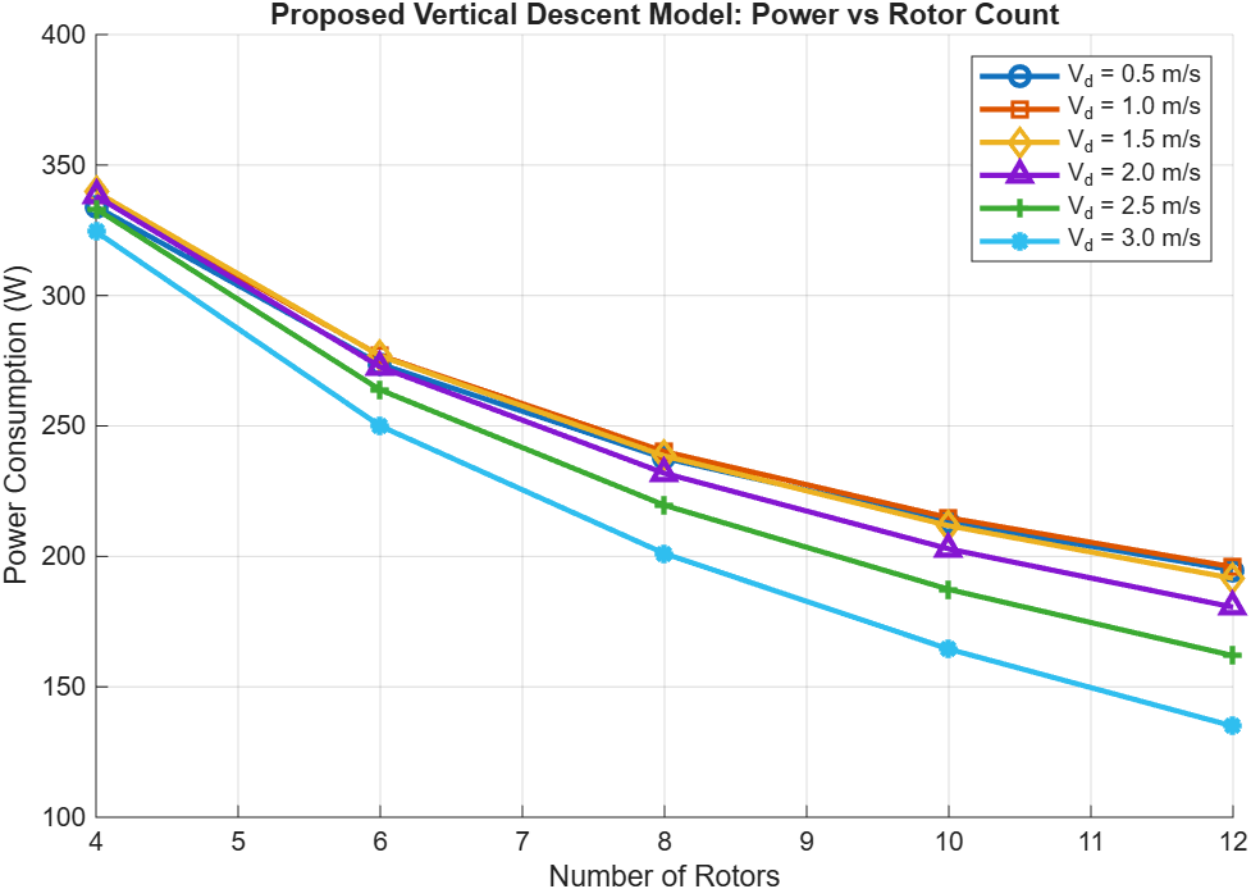


>>> APM model parameter value

Parameter	FittedValue

```
{'k1'} 0.92227
{'c2'} 1
{'c4'} 0.1
{'c5'} 0.05
```

Average Relative Difference (APM Descent Model): 5.17 %



$V_d = 3.0$ m/s: Power change from $n=4$ to $n=12$ = -58.42 % (decrease)

$V_d = 3.0$ m/s: Power change from 0.1kg to 1.0kg payload = 49.41 % (increase)